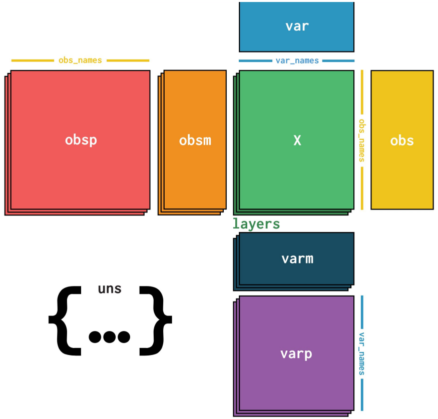


anndataR

easily interact with anndata in R

Converting single cell transcriptomics data between different frameworks is a horrible experience. 🫠

Background

- Data structures and file formats
- Cross-language interoperability
- Do we need interoperability?

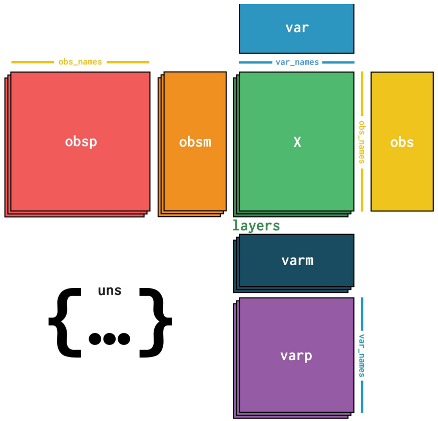
Pre-existing solutions

anndataR

- Goals
- Example

Background: the frameworks

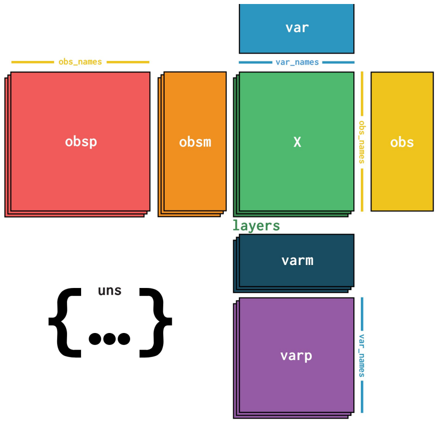
anndata



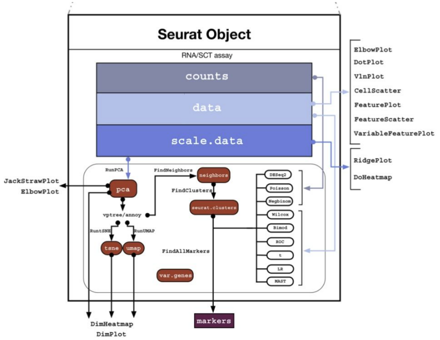
Data structure used in Python

Background: the frameworks

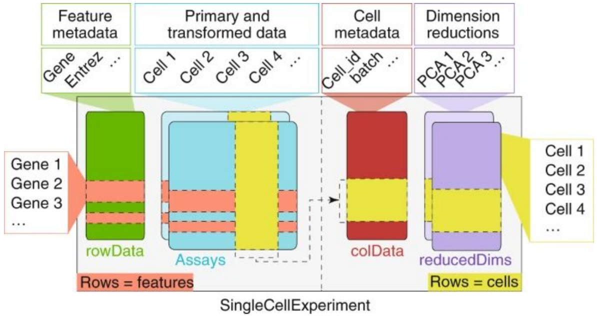
anndata



SeuratObject



SingleCell Experiment

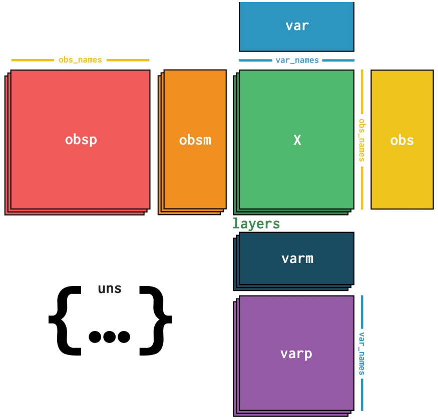


Data structure used in Python

Data structures used in R

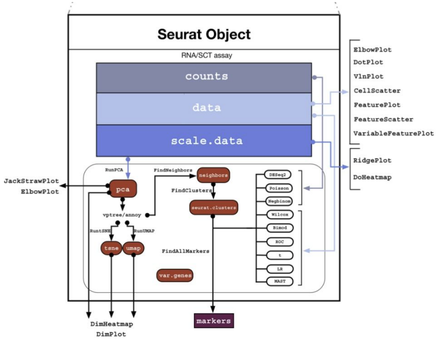
Background: the frameworks

anndata



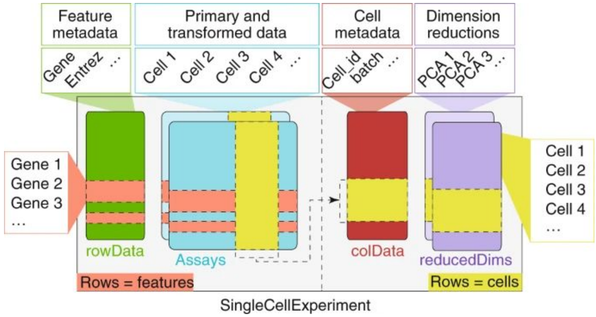
`.h5ad` or `.zarr` files

SeuratObject



`.rds` files

SingleCell Experiment



`.rds` files

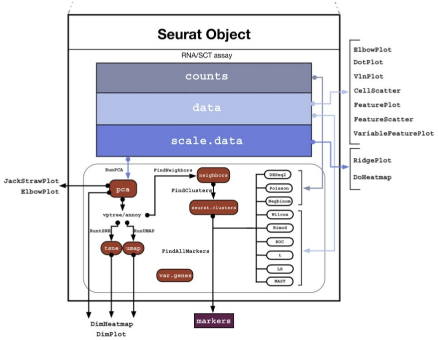
Background: the frameworks

rds files

serialised R objects

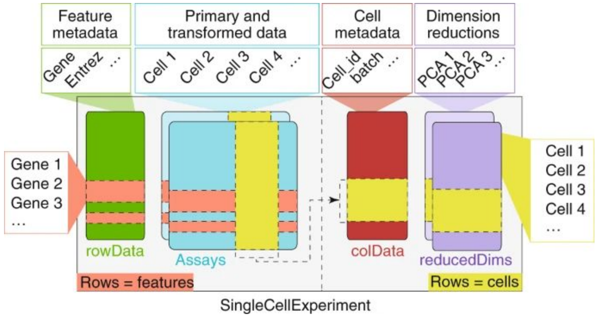
not interoperable with Python

SeuratObject



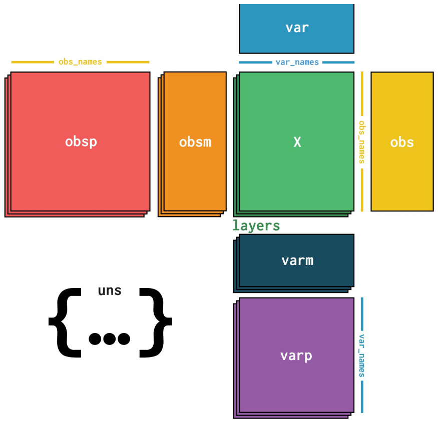
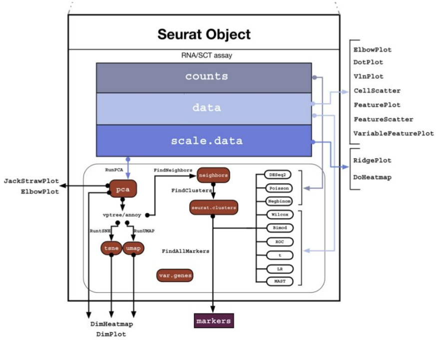
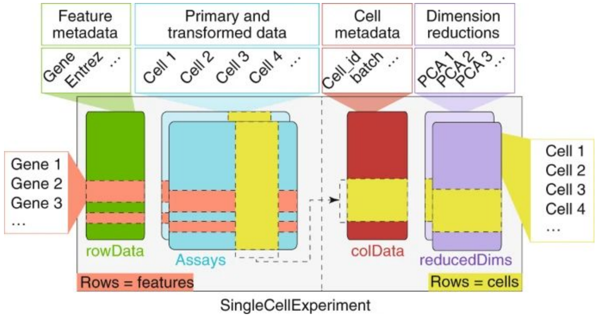
`.rds` files

SingleCell Experiment



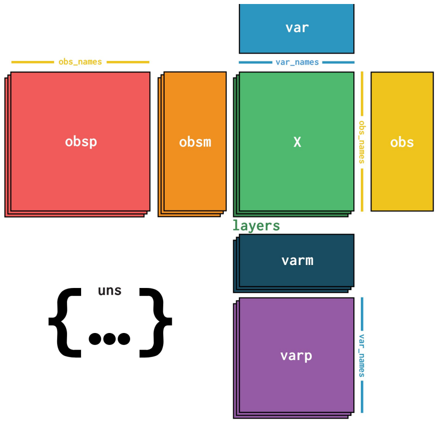
`.rds` files

Background: the frameworks

anndata	SeuratObject	SingleCell Experiment
		
`.h5ad` or `.zarr` files	`.rds` files	`.rds` files

Background: the frameworks

anndata



`.h5ad` or `.zarr` files

h5ad files

based on hdf5

libraries in many languages

one file == one object

zarr files

maintainable subset

designed for cloud storage

one folder == one object

standardised on-disk format for anndata

Background: cross-language interoperability

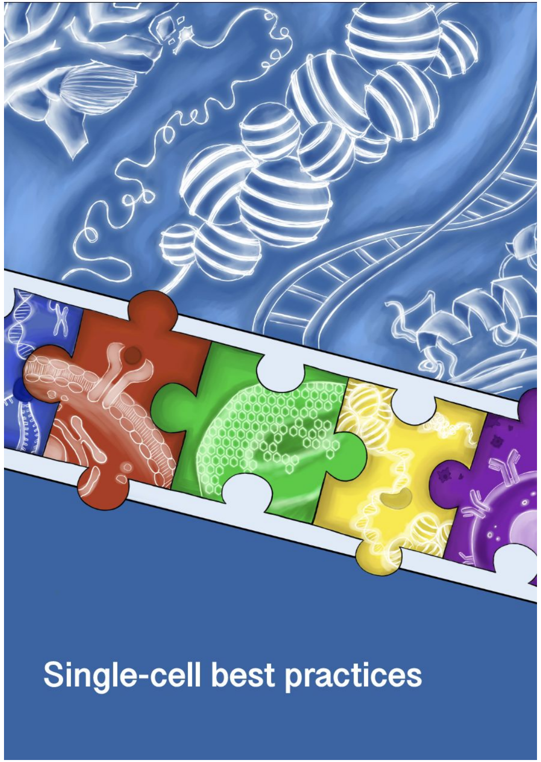
Call Python from R:

reticulate (basilisk)

Call R from Python:

rpy2

Background: do we need interoperability?



But I'm pretty sure a package for that exists already

But I'm pretty sure a package for that exists already

tool	backend	converts
theislab/zellkonverter	reticulate	AnnData ↔ SCE
dynverse/anndata	reticulate	AnnData
cellgeni/sceasy	reticulate	AnnData ↔ SCE & Seurat
ilia-kats/MuData	rhdf5	MuData ↔ SCE
PMBio/MuDataSeurat	hdf5r	MuData ↔ Seurat
mojaveazure/SeuratDisk	hdf5r	h5ad ↔ h5Seurat
mtmorgan/h5ad	rhdf5	AnnData
neurogenomics/scKirby	Other packages	Anything to anything

But we need another one!

The existing packages:

convert h5ad files to SingleCellExperiment objects and vice versa

→ information loss

read and write h5ad files using reticulate

→ difficult to use

But we need another one!

we really shouldn't be saving files as .rds files for interoperability

h5ad (& zarr) is standardised & used

→ so we should make it natively useable in R as well

anndataR features

We want anndataR to:

1. mimic anndata interface
2. use an R native HDF5 (or zarr) library (no Python needed)
3. convert to SingleCellExperiment & Seurat
4. have a maintainable design

anndataR showcase

```
> library(anndataR)

> adata <- read_h5ad(h5ad_path, to = "HDF5AnnData")
> adata

#> AnnData object with n_obs × n_vars = 50 × 100
#>     obs: 'Float', 'FloatNA', 'Int', 'IntNA', 'Bool', 'BoolNA', …
#>     var: 'String', 'n_cells_by_counts', 'mean_counts', …
#>     obsm: 'X_pca', 'X_umap'
#>     varm: 'PCs'
#>     layers: 'counts', 'csc_counts', 'dense_X', 'dense_counts'
#>     obsp: 'connectivities', 'distances'
```

anndataR showcase

```
> dim(adata$X)
#> [1]  50 100



> adata$obs[1:5, 1:6]
#>         Float FloatNA Int IntNA  Bool BoolNA
#> Cell000 42.42     NaN   0    NA FALSE  FALSE
#> Cell001 42.42   42.42   1    42  TRUE     NA
#> Cell002 42.42   42.42   2    42  TRUE   TRUE
#> Cell003 42.42   42.42   3    42  TRUE   TRUE
#> Cell004 42.42   42.42   4    42  TRUE   TRUE
```

anndataR showcase

```
> sce <- adata$to_SingleCellExperiment()
> sce

#> class: SingleCellExperiment
#> dim: 100 50
#> metadata(0):
#> assays(5): X counts csc_counts dense_X dense_counts
#> rownames(100): Gene000 Gene001 ... Gene098 Gene099
#> rowData names(11): String n_cells_by_counts ... dispersions_norm
#> colnames(50): Cell000 Cell001 ... Cell048 Cell049
#> colData names(11): Float FloatNA ... log1p_total_counts leiden
#> reducedDimNames(0):
#> mainExpName: NULL
#> altExpNames(0):
```

anndataR showcase

```
> obj <- adata$to_Seurat()
> obj

#> An object of class Seurat
#> 500 features across 50 samples within 5 assays
#> Active assay: RNA (100 features, 0 variable features)
#>  2 layers present: counts, data
#>  4 other assays present: counts, csc_counts, dense_X, dense_counts
```

anndataR showcase

```
> adata <- AnnData(
    X = matrix(rnorm(100), nrow = 10),
    obs = data.frame(cell_type = factor(rep(c("A", "B"), each = 5))),
    var = data.frame(gene_name = paste0("gene_", 1:10)))

> adata

#> AnnData object with n_obs × n_vars = 10 × 10
#>     obs: 'cell_type'
#>     var: 'gene_name'
```

scverse / anndataR

Part of the scverse *ecosystem*

Documentation, features:

https://anndatar.data-intuitive.com/articles/design.html

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- Danila Bredikhin (Kundaje lab & Engreitz lab)
- Isaac Virshup (Theis lab)
- Brian Schilder (Imperial College London)
- Chananchida Sang-aram (Saeys lab)

Disclaimer:
We're still working on zarr support

Some incompatibilities between data types remain

Join our workshop on polyglot programming on Thursday!

SaeysLab

Data mining and Modelling for Biomedicine

Postdoc positions available:

https://saeyslab.sites.vib.be/en/jobs